CSE492 ENGINEERING PROJECT
PROPOSAL FORM

Date:19/02/2026 CSE 492 Registration Semester: Fall

Make sure that you have registered to the CSE492 course via OBS before filling out this form.

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CSE 492 registration type (Mark one of the following choices) :

☒ Regular ☐ Single Course (Tek Ders) ☐ Supplementary Course (Ek Ders) ☐ Extended

Project Title	Hybrid Graph Neural Networks for Financial Fraud Detection
Abstract (200 words)	Financial fraud is an evolving threat costing the global banking industry billions annually. Traditional machine learning models often fail to detect organized fraud rings because they analyze transactions in isolation, ignoring complex relational patterns. As a professional working in the banking sector (BI Developer), I have observed that detecting multi-hop fraudulent networks requires a shift from tabular analysis to relational graph structures to ensure financial integrity. This project proposes a hybrid deep learning approach combining Graph Attention Networks (GAT) with a Variational Autoencoder (VAE). The GAT component captures structural dependencies within a heterogeneous graph (Account, Transaction, Merchant nodes). Simultaneously, the VAE learns the latent distribution of "normal" behavior for unsupervised anomaly detection. This dual mechanism enables the identification of both known fraud patterns and previously unseen anomalies. The system will be validated using a comprehensive suite of financial datasets, ranging from standardized benchmarks to large-scale synthetic transaction logs, to ensure robustness across various fraudulent scenarios. The project's overall success will be judged based on the model's ability to accurately detect anomalous behavioral patterns and provide interpretable, real-time justifications for its decisions, as detailed in the evaluation criteria section.

Hardware constraints: (PC, Apple, mobile phone, other device, etc.)

Apple MacBook Pro (M1 Pro chip).

Software constraints: (Operating System, Progr. Languages, Libraries, API's, IDE's etc)

Operating System: macOS (Native) and Docker (Linux) for environment isolation.

Languages: Python 3.10+ and SQL.

Libraries: PyTorch 2.x, **PyTorch Geometric (PyG)** for graph layers, Scikit-learn, and SHAP for explainability.

IDEs: VS Code / PyCharm.

Evaluation criteria (Quality Measurements, Scalability, Error Rate, etc)

Imbalanced Data Success: Maximizing minority class detection (F1-Score and Recall).

Scalability: Testing on the large-scale PaySim dataset (6M+ transactions) to measure training efficiency.

Explainability: Human-readable justifications for flagged transactions using SHAP values.

Latency: Target sub-second inference time for near-real-time banking alerts.

Design Methodology Diagrams

☐ Use Case Diagrams ☐ Class Diagrams ☒ Sequence Diagrams ☐ State Diagrams

Impact area

☐ Health ☐ Environmental ☒ Security ☐ Social / Ethical

☐ Sustainability ☒ Productivity ☒ Other: Financial Integrity

Project Advisor Name, Surname, Signature:	Academic Advisor Name, Surname, Signature:
Tacha Serif	Tacha Serif